rdnalink Phase 0: results

EUKARYOME v2.1 long reads; protocol locked 2026-10-07. Report built 2026-10-15 by scripts/make_report.py.

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Verdicts (results under the original protocol first, section 10)
  K2  Original protocol (barrnap-cut reads removed, section 7 split): PASS
  K2  Amendment 1 windows, section 7 split: PASS
  K2  Amendment 1 windows, note 2 assembly split: PASS
  K1  Original protocol: FAIL (recall: chimeras 29.8%, swaps 13.5%; precision 16.0%)
  K1  Amendment 1 windows: FAIL (recall: chimeras 31.1%, swaps 14.9%; precision 32.0%)
  K2  at matched known-kept rates (note 7): not supported; no shared level in 4 of 4 groups
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K2 passes if, at 90% known-kept (or the highest level all methods reach), R = 1 - FK(L)/FK(best of B1, B2) >= 0.25 in at least two of Agaricales, Hypocreales and Rhabditida, and FK(L) <= FK(best) in Glomeromycota (species level, per-read rates). K1 passes if spiked chimeras and label swaps are each caught >= 80% of the time (pooled) and >= 50% of a random sample of real flags hold up on trees.

K2, original protocol: false known (FK, % of reads) at the known-kept level, per read, species level

group	at KK %	FK B1	FK B2	FK L	R	meets rule
Glomeromycota	0	2.3	0.0	0.0	-	FK(L) <= best: yes
Agaricales	0	3.3	8.4	0.0	1.00	R >= 0.25: yes
Hypocreales	3	13.2	8.2	0.0	1.00	R >= 0.25: yes
Rhabditida	41	2.6	7.9	5.8	-1.20	R >= 0.25: no

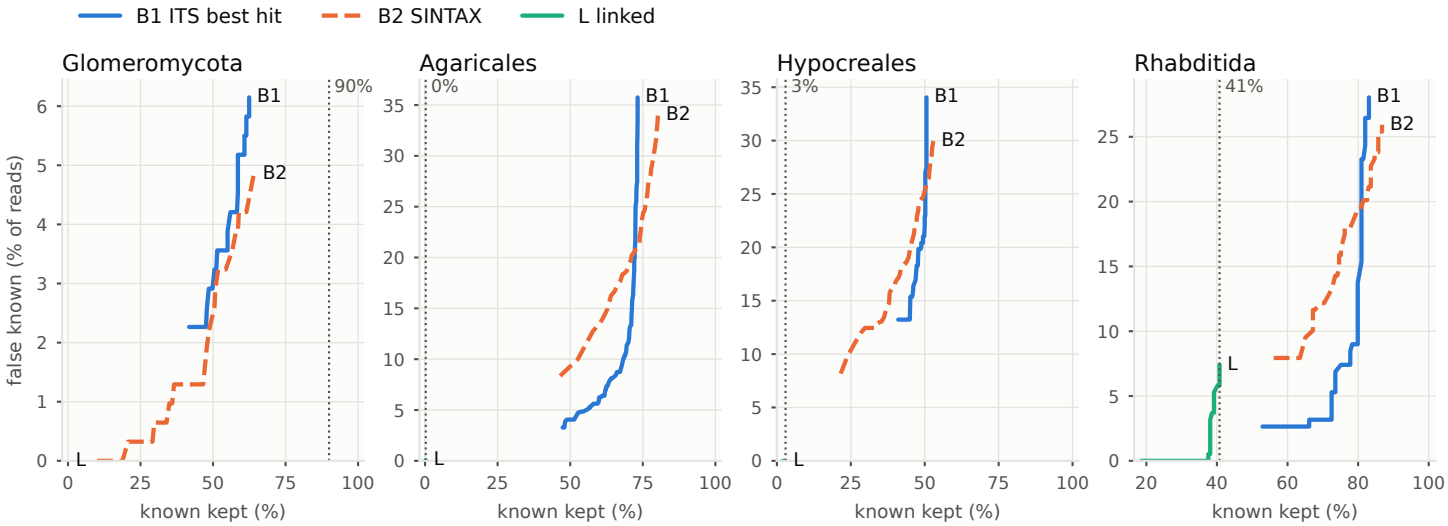


Figure. Original protocol, species level, per read: false known against known kept as the threshold (q for B1 and L, c for B2) is swept; dotted line, the level K2 is read at.

Original protocol, rates averaged over species (macro; reported, not deciding)

group	at KK %	FK B1	FK B2	FK L	R	meets rule
Glomeromycota	0	2.4	0.0	0.0	-	FK(L) <= best: yes
Agaricales	0	3.5	9.4	0.0	1.00	R >= 0.25: yes
Hypocreales	3	14.8	7.9	0.0	1.00	R >= 0.25: yes
Rhabditida	46	1.9	13.1	8.1	-3.25	R >= 0.25: no

K2 with the amendments

```
Amendment 1 windows, section 7 split: PASS
group           at KK %   FK B1   FK B2    FK L       R          meets rule
Glomeromycota         0     2.3     0.0     0.0       -  FK(L) <= best: yes
Agaricales            0     3.3     8.4     0.0    1.00      R >= 0.25: yes
Hypocreales           2    13.2     8.2     0.0    1.00      R >= 0.25: yes
Rhabditida           40     2.5     6.5     5.5   -1.20       R >= 0.25: no

Amendment 1 windows, note 2 assembly split: PASS
group           at KK %   FK B1   FK B2    FK L       R          meets rule
Glomeromycota         0     2.3     0.0     0.0       -  FK(L) <= best: yes
Agaricales            0     3.3     8.4     0.0    1.00      R >= 0.25: yes
Hypocreales           1    13.2     8.2     0.0    1.00      R >= 0.25: yes
Rhabditida           40     2.5     6.5     5.5   -1.20       R >= 0.25: no
```

K1

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K1, Original protocol: FAIL
group                chimeras caught          swaps caught   real F1   real F2
Glomeromycota        157/399 (39.3%)          2/200 (1.0%)         3         0
Agaricales             31/400 (7.8%)          0/200 (0.0%)        96         0
Hypocreales            46/400 (11.5%)         5/200 (2.5%)        40         0
Rhabditida           225/342 (65.8%)         99/183 (54.1%)        2         0
all                 459/1541 (29.8%)        106/783 (13.5%)      141         0
Precision on trees (real/checked): F1 8/50 (16.0%); F2 0/0 (-%); all 8/50 (16.0%)

K1, Amendment 1 windows: FAIL
group                chimeras caught          swaps caught   real F1   real F2
Glomeromycota        156/400 (39.0%)          2/200 (1.0%)         7         0
Agaricales             32/400 (8.0%)          0/200 (0.0%)       115         0
Hypocreales            44/400 (11.0%)         1/200 (0.5%)        43         0
Rhabditida           266/400 (66.5%)        116/200 (58.0%)        2         0
all                 498/1600 (31.1%)        119/800 (14.9%)      167         0
Precision on trees (real/checked): F1 16/50 (32.0%); F2 0/0 (-%); all 16/50 (32.0%)
```

Real flags are reads whose markers conflict on per-marker trees. EUKARYOME documents a per-marker BLASTn check for its (meta)genome and (meta)transcriptome data only, so they are not described as chimeras its checks missed (note 4). Their accessions are in scores/k1_*/flags.tsv.

Checks and exclusions

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Depth check (note 2: a group with more than 1% of fold queries failing reads ranks <= 1,000)
  Glomeromycota: 2721 of 5620 fold queries fail (48.42%); ranks <= 1000
  Agaricales: 628 of 10598 fold queries fail (5.93%); ranks <= 1000
  Hypocreales: 214 of 3602 fold queries fail (5.94%); ranks <= 1000
  Rhabditida: 0 of 2003 fold queries fail (0.00%); ranks <= 300
Fold calls whose 0.2-point set reaches the last rank read (note 2): 0
```

Reads and windows: 400,217 reads; 397,441 cut into three windows, 31,563 of them by barrnap (removed under the original protocol).
Not cut: missing_LSU 1,296; missing_SSU 684; strand_or_order 439; missing_SSU_LSU 344; multi_operon 13.
Query reads per group: Glomeromycota 5,546; Agaricales 9,436; Hypocreales 3,318; Rhabditida 2,901.

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Protocol, amendments and notes, with the dates they were locked
  AMENDMENT1_note1.md (locked 2026-10-10): Phase 0 protocol, Amendment 1, note 1: ITSxRust and barrnap run in
      chunks
  AMENDMENT1_windows.md (locked 2026-10-08): Phase 0 protocol, Amendment 1: how the marker windows are cut
  NOTE2_search.md (locked 2026-10-14): Phase 0 protocol, note 2: the search, the folds, the K1 spike-ins and the
      scoring definitions
  NOTE3_k1_trees.md (locked 2026-10-14): Phase 0 protocol, note 3: how the K1 tree check reads a tree
  NOTE4_eukaryome_checks.md (locked 2026-10-14): Phase 0 protocol, note 4: what EUKARYOME's own checks cover,
      and how the report describes real K1 flags
  NOTE5_code_fixes.md (locked 2026-10-14): Phase 0 protocol, note 5: code fixes before scoring
  NOTE6_sintax_fix.md (locked 2026-10-15): Phase 0 protocol, note 6: a code fix during the SINTAX run
  NOTE7_reading_k2.md (locked 2026-10-15): Phase 0 protocol, note 7: reading K2 when no known-kept level is
      shared
  PROTOCOL_PHASE0.md (locked 2026-10-07): Phase 0 protocol: do linked SSU-ITS-LSU calls beat ITS alone outside
      AMF?
```

Every number in this report is in numbers.tsv, with the file it came from.

After scoring (note 7)

Added after scoring (note 7). These analyses were written once the results were known; they change no number in the pages before, and are read from the same score tables.

Reading K2 at matched known-kept rates

Section 8 reads false known at 90% known kept, or at the highest known kept all methods reach, so that methods are compared at matched known-kept rates (section 6). At species level, the known-kept range of L lies wholly below those of B1 and B2 in every group (table): no level is reached by all three, so no matched comparison exists. score_k2.py then reads each baseline at its own lowest known kept and L at its highest, and k2_decision.txt reports PASS from these unmatched values. Read at matched rates, no group meets section 8, and K2 is not supported. L stays this low at every q, so its threshold is not the limit: L names a species only when SSU, ITS and LSU each give the same unique best species, and the SSU and LSU rarely give the same one as the ITS.

Known-kept range (%) of each method over its sweep, species level, per read, original protocol

group	L	B1	B2	shared level
Glomeromycota	0.0-0.0	41.7-62.5	10.0-64.4	none
Agaricales	0.1-0.2	47.4-73.2	46.5-80.2	none
Hypocreales	1.9-2.9	41.1-50.6	21.6-52.9	none
Rhabditida	18.5-40.7	52.9-83.1	56.1-86.8	none

Why K1 misses

K1 misses on recall in every group. A chimera is flagged when two markers name different genera with confidence. The SSU does so for few fungal spike-ins, so chimeras that join the SSU of one read to the ITS and LSU of another (chim_b) are almost never caught in fungi; in Agaricales and Hypocreales the LSU is also rarely confident, which limits chimeras that swap only the LSU (chim_a). Of the 50 sampled real flags checked on trees, 27 had at least two markers placed in a supported clade, the minimum for a flag to be real.

Spike-ins caught, by kind, and share of spike-ins on which each marker names a genus with confidence (original protocol)

group	chim_a	chim_b	swaps	confident: SSU / ITS / LSU
Glomeromycota	152/199	5/200	2/200	1% / 92% / 82%
Agaricales	30/200	1/200	0/200	0% / 78% / 20%
Hypocreales	39/200	7/200	5/200	6% / 73% / 22%
Rhabditida	125/172	100/170	99/183	60% / 89% / 76%